Phase2_Daily_Schedule

Phase 2 — Day-by-Day Schedule · Biomedical Imaging (core + expansion + video)

Weeks 13–30 · Mon Nov 30, 2026 → Sun Apr 4, 2027 · ~385 hrs (+ Capstones 1 & 2) · → Tier 0/1

Goal: own MRI / fluorescence / whole-slide segmentation end-to-end, then extend into phenomics, atlas registration, WSI engineering, and pathology foundation models; finish with video/temporal. Two shippable capstones.

Blocks: A 06:00–08:00 (theory) · **B** 08:30–10:30 (build + threads) · **Evening** 20:30–22:30 (papers, Anki, R). **Weekend** = ~10 h buffer + one rest day. **Threads (running): T** DSA+C++ (first ~45 min of Block B) · **M** implement-from-scratch (Block B) · **R** reproduce-a-paper + write + apply (Evening, ~2 h/wk).

Block 2A — Segmentation core (Weeks 13–22)

Week 13 · Nov 30–Dec 6 — Imaging physics

Day	Block A	Block B (T/M + build)	Evening (+ R)
Mon	MRI contrast: T1/T2/FLAIR/T1CE (MRI Q&A)	**T:** trees/BFS-DFS → load & view public MRI (nibabel)	OHBM MRI physics; Anki
Tue	MRI artifacts: motion, bias field, partial volume	**M:** bias-field correction demo (N4/SITK)	Callaghan L1–2; **R:** pick imaging paper
Wed	Fluorescence: photobleaching, bleed-through, autofluorescence (iBiology)	**T:** heaps → inspect a fluorescence stack	Anki; glossary
Thu	k-space & acquisition basics	**M:** simple intensity normalization pipeline	Callaghan L3–4
Fri	Preprocessing decisions that affect model design	**T:** intervals/greedy	Whiteboard-Fri: which artifacts to correct & why; **R:** write-up
Sat–Sun	**Buffer + rest** · *Deliverable:* 1-page physics note (T1/T2/FLAIR/T1CE + fluorescence artifacts)		

Week 14 · Dec 7–13 — CNNs, U-Net family, convolution math

Day	Block A	Block B	Evening
Mon	CS231n L5: CNNs; conv arithmetic (Dumoulin)	**T:** DP intro → implement conv2d from scratch	Anki
Tue	U-Net (read; derive output sizes)	**M:** hand-compute U-Net shapes; code encoder	Re-read U-Net; Anki
Wed	V-Net (3D) + Attention U-Net	**T:** DP on grids → code decoder + skips	Anki
Thu	nnU-Net paper (read twice)	**M:** finish a minimal U-Net; overfit 1 image	nnU-Net paper again; **R:** reproduce-paper
Fri	Receptive fields; why skips work	**T:** graphs/union-find	Whiteboard-Fri: U-Net from memory + shapes
Sat–Sun	**Buffer + rest**		

Week 15 · Dec 14–20 — Loss functions (the imbalance killer)

Day	Block A	Block B	Evening
Mon	Loss survey (Jadon); Dice math	**M:** Dice + BCE from scratch; unit-test	Anki
Tue	Generalized Dice; class weighting	**M:** Generalized Dice; gradient vs class-ratio plot	Anki
Wed	Focal & Tversky (severe imbalance)	**M:** Focal, Tversky, Focal-Tversky	Anki
Thu	Boundary loss (thin structures)	**M:** Boundary loss + compound DiceCE	**R:** write-up
Fri	When each loss helps	**T:** review + timed set	Whiteboard-Fri: derive Dice gradient + empty-mask instability
Sat–Sun	**Buffer + rest** · *Deliverable:* tested losses library + gradient plot		

Week 16 · Dec 21–27 — 🔴 CONSOLIDATION (holiday-light, ~18 h)

Day	Block A	Block B	Evening
Mon-Fri	Re-derive: conv output sizes, Dice gradient; re-read nnU-Net design	Re-implement (blank file): conv2d + Dice; **T:** spaced review	Anki catch-up; **R:** finish a reproduce-paper
Sat–Sun	**Rest** (holidays)		

Week 17 · Dec 28–Jan 3 — nnU-Net deep dive + MONAI

Day	Block A	Block B	Evening
Mon	nnU-Net fingerprinting → auto-config	**T:** DP/backtracking → clone & run MONAI brain-seg notebook	MONAI videos; Anki
Tue	MONAI transforms & data pipeline	**M:** build a MONAI training loop	Anki
Wed	nnU-Net v2 source read	**T:** matrix/2D problems → modify nnU-Net config	Kitware comparison
Thu	Patch-based training, sliding-window inference	**M:** sliding-window inference	**R:** reproduce-paper
Fri	Preprocessing/spacing/normalization choices	**T:** timed medium set	Whiteboard-Fri: walk nnU-Net config end-to-end
Sat–Sun	**Buffer + rest**		

Week 18 · Jan 4–10 — MONAI + microscopy; reproduce nnU-Net

Day	Block A	Block B	Evening
Mon	AI-for-Medical-Diagnosis: brain-MRI U-Net module	**T:** strings → set up BraTS-subset run	DigitalSreeni (U-Net)
Tue	Evaluation module (metrics)	**M:** train nnU-Net on BraTS subset	Anki
Wed	Microscopy-native tips (DigitalSreeni)	**T:** two-pointer/window → adapt to fluorescence	DigitalSreeni code-along
Thu	What changes MRI→fluorescence	**M:** fluorescence tiling + inference	**R:** write-up
Fri	Review core seg	**T:** timed set	Whiteboard-Fri: nnU-Net defaults, what you'd change for microscopy
Sat–Sun	**Buffer + rest** · *Deliverable:* reproduced nnU-Net + write-up		

Week 19 · Jan 11–17 — Transformer segmentation + foundation models

Day	Block A	Block B	Evening
Mon	ViT (re-derive attention)	**M:** patch embedding + a ViT block	Anki
Tue	TransUNet / Swin-UNet	**T:** graphs → run a Swin-UNet demo	Anki
Wed	UNETR / Swin UNETR (3D)	**M:** wire a UNETR-style decoder	Anki
Thu	SAM / MedSAM (promptable)	**T:** intervals → run MedSAM on a sample	**R:** reproduce-paper
Fri	MedNeXt (ConvNet that still wins)	**T:** timed set	Whiteboard-Fri: CNN vs ViT inductive biases; SAM design
Sat–Sun	**Buffer + rest**		

Week 20 · Jan 18–24 — 3D, uncertainty, evaluation

Day	Block A	Block B	Evening
Mon	TorchIO: volumetric augmentation	**M:** a 3D augmentation pipeline	Anki
Tue	MC-Dropout, deep ensembles (Yarin Gal)	**M:** add MC-Dropout + predictive entropy	Anki
Wed	Calibration (ECE, reliability)	**T:** DP → compute ECE + reliability diagram	Anki
Thu	Metrics Reloaded (choosing metrics)	**M:** proper Dice/IoU/HD95 eval	**R:** write-up
Fri	Domain shift (Stanford AIMI)	**T:** 3D-conv shape/param drill (by hand)	Whiteboard-Fri: pick metrics for a task; UQ readout
Sat–Sun	**Buffer + rest**		

Weeks 21–22 · Jan 25 – Feb 7 — 🎯 CAPSTONE 1 (public brain-seg) + self-test

Day (both weeks)	Block A	Block B	Evening (+ R)
Mon–Fri (Wk 21)	Design decisions (preprocessing→arch→loss→UQ)	Build: preprocessing + 3-architecture ablation on BraTS/Decathlon subset	Log results; **R:** capstone as public repo
Mon–Thu (Wk 22)	Analysis & failure modes	Loss ablation + UQ + failure-mode analysis; **T:** keep 1 timed set/day	Write the 3-page write-up
Fri (Wk 22)	🏁 **PHASE-2-CORE SELF-TEST** (U-Net shapes; Dice gradient; why nnU-Net wins; diagnose bias-field/imbalance/label-noise/domain-shift)	Polish repo + README	**R:** post write-up; ▶ **Tier 0/1 applications**

Sat–Sun	**Buffer + rest** · *Deliverable:* Capstone-1 notebook + write-up (public)		

Block 2B — Discovery-imaging expansion (Weeks 23–26)

Week 23 · Feb 8–14 — Cell/nucleus segmentation + phenomics (2.6)

Day	Block A	Block B	Evening
Mon	Cellpose & StarDist (instance seg)	**M:** run Cellpose + StarDist on a public cell set	Anki
Tue	Cell Painting assay + morphological features	**T:** graphs → CellProfiler feature extraction	JUMP-CP / RxRx overview
Wed	Dose–response / EC50, ROC-AUC readouts	**M:** fit a dose–response curve	Anki
Thu	Batch effects in phenomics	**T:** DP → batch-correction demo	**R:** reproduce-paper
Fri	Profiling pipeline design	**T:** timed set	Whiteboard-Fri: what a Cell-Painting profile encodes
Sat–Sun	**Buffer + rest** · *Deliverable:* segment → features → AUC readout (public)		

Week 24 · Feb 15–21 — 3D + atlas registration (2.7) + WSI engineering start (2.8)

Day	Block A	Block B	Evening
Mon	Registration: rigid/affine/deformable	**M:** ANTsPy register a mouse-brain volume	Anki
Tue	Allen atlas; BrainGlobe/brainreg	**T:** trees → atlas-based region readout	Anki
Wed	Registration error → metric propagation	**M:** per-region quantification	Anki
Thu	WSI: tiling, pyramidal I/O, zarr/dask	**T:** strings → OpenSlide tiling script	TIAToolbox docs; **R:** write-up
Fri	Keeping tile context; throughput	**T:** timed set	Whiteboard-Fri: registration types + error propagation
Sat–Sun	**Buffer + rest**		

Week 25 · Feb 22–28 — WSI finish (2.8) + stain-norm/MIL/active learning (2.9)

Day	Block A	Block B	Evening
Mon	Tiled inference + stitching (TIAToolbox)	**M:** tiled inference on a public WSI	Anki
Tue	Stain normalization (Macenko/Vahadane)	**M:** torchstain normalization + ablation	Anki
Wed	MIL for weak slide labels (CLAM)	**T:** graphs → CLAM MIL classifier	Anki
Thu	Active learning (uncertainty/diversity)	**M:** an active-learning loop (modAL)	**R:** reproduce-paper
Fri	Annotation efficiency	**T:** timed set	Whiteboard-Fri: MIL assumption + stain-norm rationale
Sat–Sun	**Buffer + rest**		

Week 26 · Mar 1–7 — Pathology foundation models & SSL (2.10) + imaging-track capstone

Day	Block A	Block B	Evening
Mon	DINOv2 / MAE (SSL objectives)	**M:** extract UNI embeddings on a public task	Virchow paper (skim)
Tue	Pathology FMs (UNI/CONCH/Virchow)	**T:** DP → linear-probe on FM embeddings	Anki
Wed	Linear-probe vs fine-tune	**M:** fine-tune head; compare to supervised CNN	Anki
Thu	When FM embeddings win (low-data)	**T:** timed → build the comparison	**R:** write-up
Fri	Wrap the imaging-track capstone	Polish repo (seg → features → FM-embeddings comparison)	Whiteboard-Fri: SSL objective; FM vs CNN
Sat–Sun	**Buffer + rest** · *Deliverable:* imaging-track capstone (public) — MIDL/MICCAI-workshop candidate		

Block 2C — Imaging & video extension (2V) (Weeks 27–30)

Week 27 · Mar 8–14 — Video & temporal models

Day	Block A	Block B	Evening

Mon	VideoMAE / TimeSformer	**M:** a temporal model on a toy clip set	Anki
Tue	ViViT / SlowFast	**T:** DP-on-sequences → run a clip classifier	Anki
Wed	SAM 2 (memory for video seg)	**M:** SAM 2 mask propagation on a clip	Anki
Thu	RAFT (optical flow)	**T:** graphs → optical-flow demo	**R:** reproduce-paper
Fri	Video-transformer vs frame+track	**T:** timed set	Whiteboard-Fri: SAM 2 memory mechanism
Sat–Sun	**Buffer + rest**		

Week 28 · Mar 15–21 — Tracking + pose

Day	Block A	Block B	Evening
Mon	Tracking-by-detection; ByteTrack	**M:** run ByteTrack on a public clip	Anki
Tue	OC-SORT / BoT-SORT; HOTA metric	**T:** intervals → compute HOTA	Anki
Wed	Pose: DeepLabCut	**M:** DeepLabCut on a sample	Anki
Thu	Pose: SLEAP	**T:** graphs → SLEAP run	**R:** write-up
Fri	ID-switching & re-ID	**T:** timed set	Whiteboard-Fri: tracking-by-detection vs joint
Sat–Sun	**Buffer + rest**		

Weeks 29–30 · Mar 22 – Apr 4 — Cell tracking/lineage + 🎯 CAPSTONE 2 + self-test

Day	Block A	Block B	Evening (+ R)
Mon–Fri (Wk 29)	Cellpose→TrackMate/Ultrack; lineage	Build Capstone 2 (pose→behavior **or** cell-track→lineage); SAM 2 for masks; **T:** 1 timed set/day	Log; **R:** capstone repo
Mon–Thu (Wk 30)	Lineage failure modes; eval	Finish + evaluate + write-up; **T:** keep timed sets	Write-up
Fri (Wk 30)	🏁 **PHASE-2V SELF-TEST** (SAM 2 memory; tracking-by-detection vs joint; when a video transformer wins; lineage failure modes)	Polish repo + README	**R:** post; ▶ **Tier 1 applications** (Recursion/PathAI)
Sat–Sun	**Buffer + rest** · *Deliverable:* Capstone 2 (public)		

End of Phase 2 → Phase 3 (DL theory) next. You now hold Tier 0/1: segmentation + phenomics + video, two public capstones, threads live.